

# Average Formula Handbook (Interview Level)

## 1. Basic Average

Average = Sum of observations / Number of observations. Sum = Average  $\times$  Number.

## 2. Missing Value Problems

Missing value = (Required Average  $\times$  Total Numbers) – Sum of Known Values.

## 3. Replacement of Members

New Average = Old Average + (New Value – Old Value)/n. Replaced Value = New Value – n(New Avg – Old Avg).

## 4. Consecutive Numbers

Average of consecutive numbers = Middle number. Sum = Average  $\times$  Count.

## 5. Weighted Average

Weighted Average =  $(\sum wx)/(\sum w)$ .

## 6. Combined Average

Combined Average =  $(n_1A_1 + n_2A_2)/(n_1+n_2)$ .

## 7. Average Speed

Equal distances: Average Speed =  $2xy/(x+y)$ . General: Total Distance/Total Time.

## 8. Average Marks & Scores

New Average after correction = Old Average + (Correct – Wrong)/n. Excluded Value =  $nA_1 - (n-1)A_2$ .

## 9. Mixture & Average

Average concentration = Total quantity of ingredient / Total quantity of mixture  $\times 100\%$ .

## 10. Interview Shortcuts

New person's value =  $(n+1)A_2 - nA_1$ . Leaving person's value =  $nA_1 - (n-1)A_2$ . Combined sum = Average  $\times$  Count.